

NVIDIA Research | March 2026



Nemotron-Cascade 2

Post-Training LLMs with Cascade RL and Multi-Domain On-Policy
Distillation



30B MoE



3B Active



Gold Medal



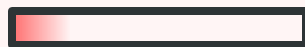
IMO 2025



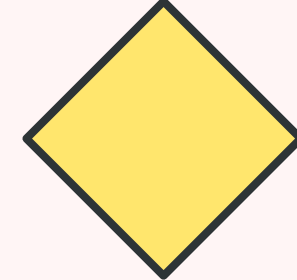
IOI 2025



ICPC 2025



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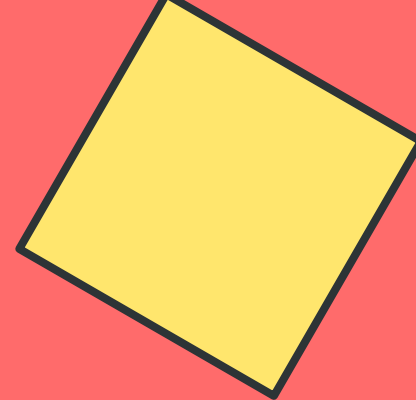
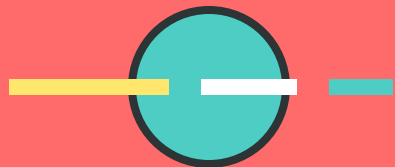
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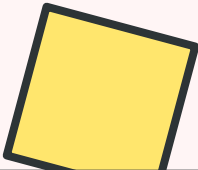
Chapter 01

Introduction & Overview

Understanding Nemotron-Cascade 2's breakthrough in efficient LLM
post-training



Model Overview & Key Achievements



Model Architecture

Total Parameters	30B
Activated Parameters	3B
Architecture	Mixture-of-Experts
Base Model	Nemotron-3-Nano-30B-A3B



Historical Significance

- Second Open-Weight LLM to achieve gold-medal performance in both IMO and IOI, following DeepSeek-V3.2-Speciale-671B-A37B
- 20× Fewer Parameters than frontier models while delivering competitive performance, demonstrating remarkable intelligence density
- Best-in-Class Performance across mathematics, code reasoning, general alignment, and instruction following benchmarks



Gold Medal Achievements

- IMO 2025: 35/42 points
- IOI 2025: 439.28/600 points
- ICPC WF 2025: 10/12 problems

Open Source Release

Model weights, training data, and methodology fully available



Technical Innovations

1

Expanded Cascade RL

Substantially expanded Cascade RL to cover a **much broader spectrum** of reasoning and agentic domains compared to Nemotron-Cascade 1.

Mathematical Reasoning

Code Generation

Agentic Tasks

Tool Use

2

Multi-Domain On-Policy Distillation

Introduced **MOPD** to distill knowledge from the strongest intermediate teacher models for each domain throughout the Cascade RL process.

- ✓ Efficiently recover benchmark regressions
- ✓ Sustain strong performance gains
- ✓ Unify specialized expertise into cohesive policy

3

Performance Recovery Mechanism

MOPD serves as a **critical stabilization point** in the training pipeline, effectively recovering performance that may regress during specialized RL stages.

3

Domain Teachers

52

Training Steps



Key Insight

By distilling from the best-performing intermediate checkpoints for each domain, MOPD creates a **balanced and robust final policy** that maintains capabilities across all benchmarks.



Faster convergence than GRPO alone

Cascade RL Framework Benefits



Catastrophic Forgetting Resistance

Domain-specific RL stages are **remarkably resistant** to catastrophic forgetting. They rarely degrade benchmark performance attained in earlier domains and may even improve it.

Preserves earlier capabilities while learning new domains

Granular control over domain interactions



Domain-Specific Optimization

RL hyperparameters and training curriculum can be **carefully tailored** to each specific domain, enabling optimized learning dynamics and improved final performance.

Customized learning rates per domain

Specialized reward functions



Compute Efficiency

Task homogeneity within each RL stage yields **substantial compute savings**, as response lengths and verification wall-clock times are more uniform within a domain.

Uniform response lengths per stage

Optimized verification throughput



Seven sequential stages: IF-RL → Multi-domain RL → MOPD → RLHF → Long-context RL → Code RL → SWE
RL





Chapter 02

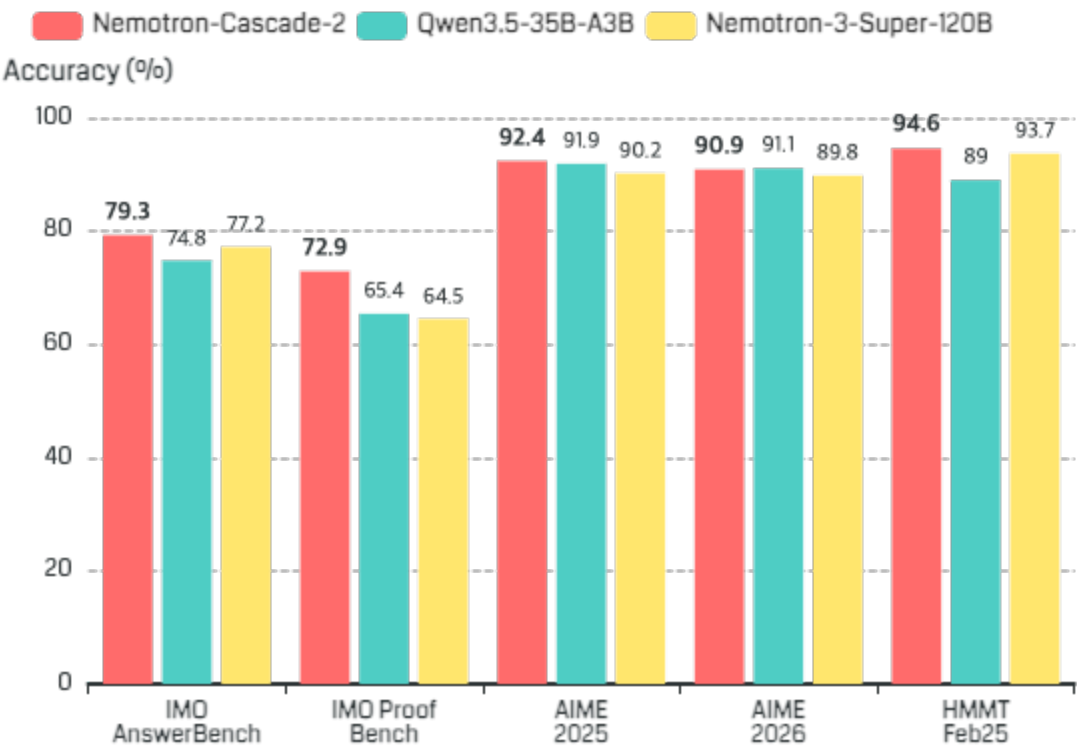
Main Results & Benchmarks

Comprehensive performance evaluation across multiple domains



Mathematical Reasoning Performance

Benchmark Results Comparison



IMO 2025 Gold Medal

35/42

Points

Gold

Medal

Solved 5 out of 6 problems, including full marks on P1-P5

Key Highlights

- ✓ AIME 2025: 92.4% Pass@1 (98.6% with TIR)
- ✓ AIME 2026: 90.9% Pass@1 (95.0% with TIR)
- ✓ HMMT Feb 2025: 94.6% Pass@1
- ✓ IMO Proof Bench: 72.9% overall

Outperforms Qwen3.5-35B-A3B and Nemotron-3-Super-120B

Code Reasoning & Competitive Programming



IOI 2025 Gold Medal

439.28

/600 Points

Gold

Medal

10/12

Full Score

Full score on P3 and P4, 86+ points on P2 in just 5 rounds using test-time scaling



ICPC World Finals 2025

Problems Solved

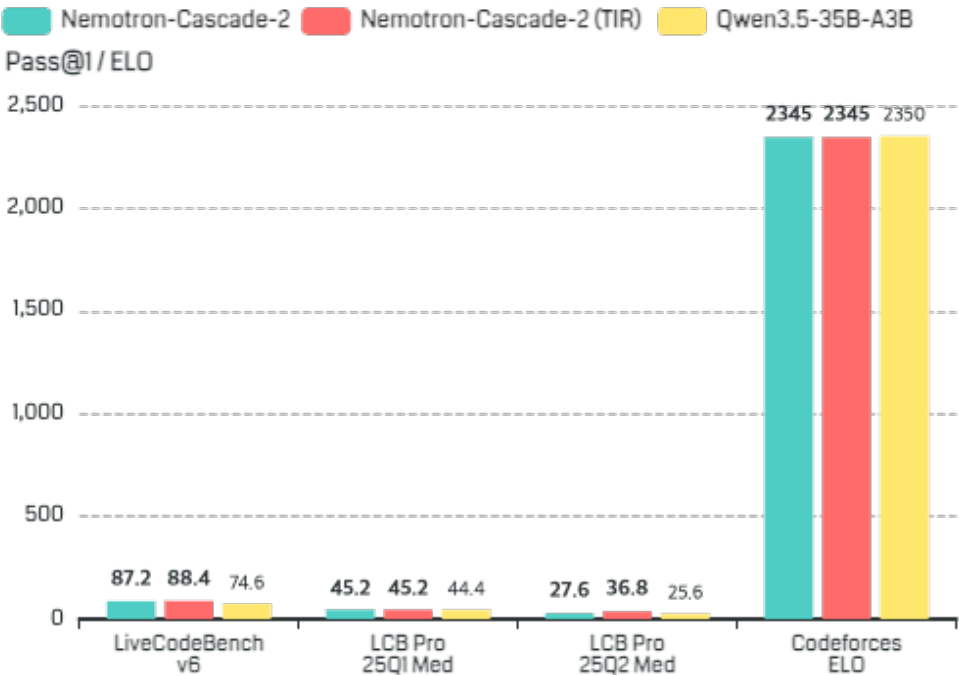
10/12

Placement

#4 Gold

8 problems solved within 100 submissions (except A and I)

Competitive Coding Benchmarks



2345

Codeforces ELO

99.7%

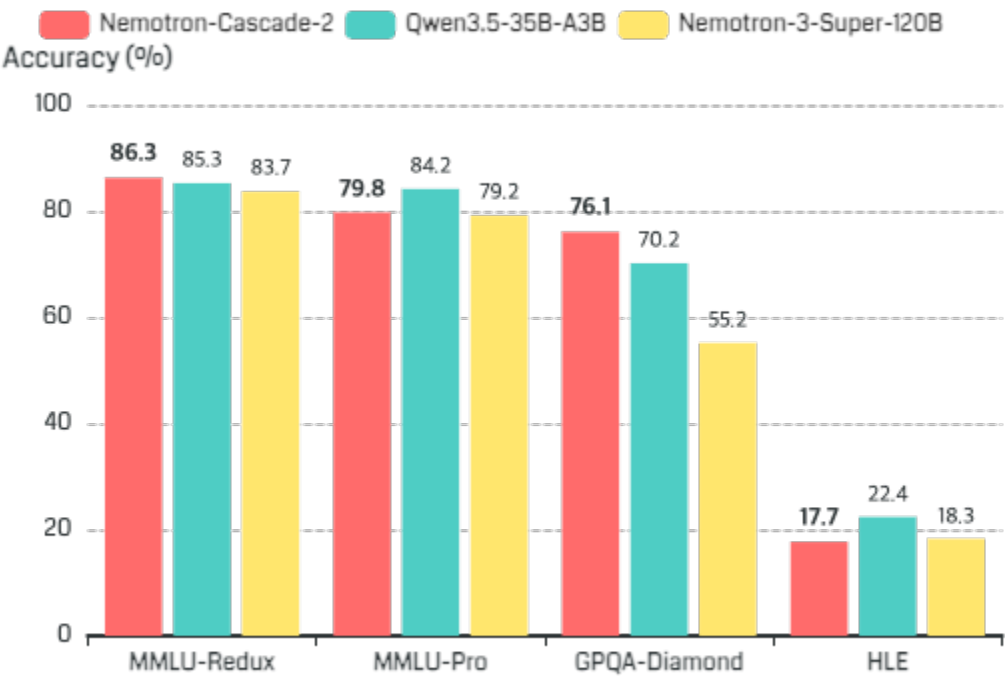
Percentile



With Tool-Integrated Reasoning (TIR): LiveCodeBench v6 reaches 88.4%, matching models with 300B+ parameters

Knowledge, STEM & Alignment Benchmarks

Knowledge & STEM Performance



86.3%

MMLU-Redux

79.8%

MMLU-Pro

76.1%

GPQA-Diamond

17.7%

HLE (no tool)

Alignment & Instruction Following

ArenaHard v2

83.5%

Human-preference alignment benchmark

IFBench (prompt)

82.9%

Best-in-class instruction following

Scale AI Multi-Challenge

45.3%

Multi-turn conversation evaluation



Best-in-Class IF Performance

State-of-the-art 83.13% on IFBench with verifiable instruction following

Long Context & Agentic Capabilities

1M Context Window

99.0%

NIAH@1M

256K

Max Training

Exceptional long-context understanding with RULER benchmark subset

Long Context Benchmarks

AA-LCR

82.9%

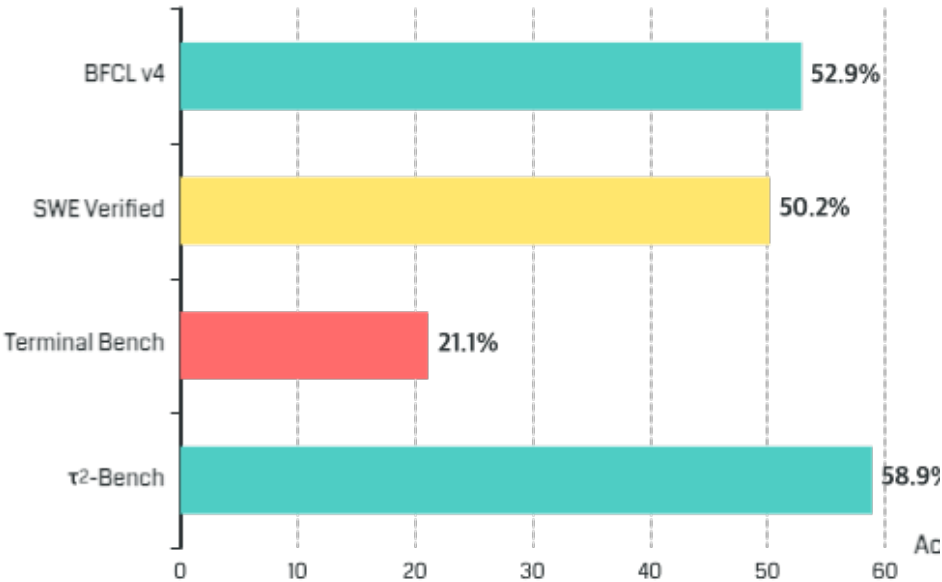
LongBench v2

39.1%

CL-Bench

12.2%

Agentic Task Performance



BFCL v4: 52.9% - Function calling across multiple languages

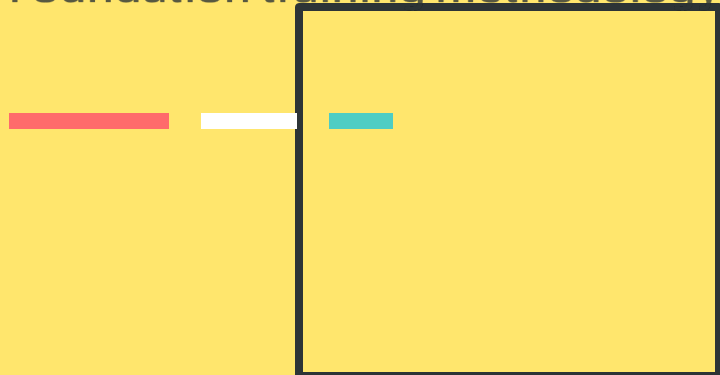
SWE Verified: 50.2% - Software engineering with OpenHands

Terminal Bench 2.0: 21.1% - End-to-end terminal workflows

Chapter 03

Supervised Fine-Tuning Framework

Foundation training methodology and chat template design



SFT Training Framework Overview



Comprehensive Data Coverage

SFT data spans a **broad range of domains** to equip the model with foundational capabilities in reasoning, conversation, instruction following, and agentic skills.



Mathematics



Coding



Science



Tool Use



Agentic



SWE



General Capabilities

Multi-turn Dialogue

Conversational flow

Knowledge QA

Question answering

Creative Writing

Story generation

Role-playing

Character acting

Safety

Harmful content

Instruction Following

Constraint adherence



Training Configuration

Sequence Length

256K

Training Epochs

~1.5

Training Steps

33K

Learning Rate

5e-5 → 5e-6



Key Design Decisions



Single-stage training with all domains mixed together



Sample packing into long sequences for efficiency



Optimal convergence at approximately 1.5 epochs



AdamW optimizer with cosine learning rate schedule

Chat Template Design



Key Innovations

1

Simplified Tags

Removed /think and /no_think tags for simplicity compared to Nemotron-Cascade 1

2

Non-Thinking Mode

Prepended empty block to activate non-thinking mode by default

3

Tool Calling

Tools specified in system prompt within tags, calls wrapped in tags

</> Template Structure

```
<lim_start> system
```

```
<lim_start> user <lim_end>
```

```
<lim_start> assistant
```

```
→ non-thinking
```

```
... → thinking
```



Tool Calling Format

System Prompt

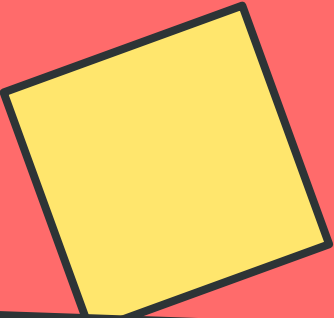
...

Tool Call

...



Adjacent tokens indicate non-thinking mode; single followed by newline indicates thinking mode



Chapter 04

SFT Data Curation

Meticulous dataset construction across ten specialized domains



Mathematics & Code Data Curation



Mathematics SFT Data

Tool-Calling Samples

1.8M

Python execution for math problems

Non-Tool Samples

2.6M

Direct reasoning without tools

Proof Samples

816K

410K generation + 400K verification

Sources: Nemotron-Cascade, Nemotron-Math-v2, Nemotron-3-Nano, AOPS split



Code Reasoning SFT Data

Unique Prompts

165K

From Codeforces, AtCoder, AIZU, CodeChef

Python Traces

1.9M

C++14 Traces

1.0M

Python Tool-Calling

1.3M



Teacher Models



DeepSeek-V3.2 - Tool-calling math



DeepSeek-V3.2-Speciale - Proofs



GPT-OSS-120B - Generation-selection



Data Processing

Deduplication

Sample I/O fingerprinting + n-gram analysis

Correctness Filtering

Only retain traces generating correct code

Result

24.2% of prompts removed as duplicates



Scientific Coding: 1.1M samples across biology, material science, physics, chemistry, mathematics

Science, Long Context & General Chat Data



Science

Total Samples 2.7M

Nemotron-Cascade 1.4M

Nemotron-3-Nano 1.3M

Domains: Physics, Chemistry, Biology

Teacher: GPT-OSS-120B



Long Context

Total Samples 234K

Nemotron-3-Nano 160K
128K avg length

ChatQA-2 74K
29K avg length

Max Sequence: 160K tokens



General Chat

Total Samples ~10M

Reasoning-on 4.9M

Reasoning-off 372K

Multi-turn 700K

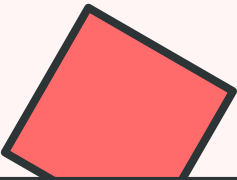
Nemotron-3-Nano 4.6M



Multi-turn conversations synthesized using two GPT-OSS-120B instances in role-playing setup



Instruction Following, Safety & Agent Data



Instruction Following

Total Samples

791K

Nemotron-Cascade 1
230K on + 64K off

Nemotron-3-Nano
457K on + 40K off

Teachers: GPT-OSS-120B, Qwen3-235B variants, DeepSeek-V3-0324



Safety

Total Samples

4K

Objective: Enable appropriate refusal behavior for unsafe inputs

Sources: Nemotron Content Safety v2, Gretel Safety, Harmful Tasks, Red-Team-2K



Conversational Agent

Tool-Use Samples

822K

Multi-turn with multiple tools

Teachers: Qwen3-235B variants, GPT-OSS-120B



Software Engineering

125K

Agentic

389K

Agentless

Scaffolds: OpenHands, SWE-Agent, Mini-SWE-Agent, Agentless



Terminal Agent

Total Samples

490K

Adapted (Math/Code/SWE)

226K

Seed-based synthetic

120K

Skill-based synthetic

140K

Chapter 05

Cascade RL Training Pipeline

Sequential domain-wise reinforcement learning stages



Cascade RL Framework & Ordering Rationale

Seven-Stage Cascade RL Pipeline

1	IF-RL	Instruction Following
2	Multi-domain RL	MCQA + Tool Calling + Structured Output
3	MOPD	Multi-Domain On-Policy Distillation
4	RLHF	Human Preference Alignment
5	Long-context RL	Extended Sequence Reasoning
6	Code RL	Competitive Programming
7	SWE RL	Software Engineering Agent



Ordering Principles



Mitigate Interference

Minimize negative interference across domains



Scale via Integration

Multi-domain RL for non-conflicting tasks



Stabilization

MOPD recovers regressed capabilities



Key Insight

The optimal ordering is a **dynamic function** of the model's behaviors and learning trajectories, not a universal constant.

SFT quality and RL environment complexity fundamentally alter behavioral dynamics, requiring different ordering than Nemotron-Cascade 1

GRPO Training Configuration

Group Relative Policy Optimization

Strict On-Policy Training

Importance sampling ratio = 1

No KL Divergence Term

Simplified REINFORCE objective

Group-Normalized Rewards

Token-level loss computation

Implementation: Nemo-RL repository

Training Dynamics

Stability

On-policy setup contributes to stable RL training and mitigates entropy collapse

Efficiency

No KL term simplifies computation while maintaining performance

Verification

Rewards verified against ground-truth answers in RLVR

GRPO Objective

$$J_{\text{GRPO}}(\theta) = E_{(q,a) \sim D}[\dots]$$

Advantage: $\hat{A}_{i,t} = (r_i - \text{mean}(\{r_i\})) / \text{std}(\{r_i\})$

-  G rollouts per prompt
-  Single gradient update per iteration
-  Policy matches data collection

Advantages

- 1 Stable training with on-policy data collection
- 2 Simplified objective without KL divergence
- 3 Group normalization reduces variance
- 4 Token-level loss for fine-grained updates

Instruction-Following RL (IF-RL)



State-of-the-Art Result

83.13%

IFBench Accuracy

Significantly improves instruction adherence through verifiable reward training



Dataset

Same instruction-following data from **NVIDIA Nano-v3** post-training

Design: Objective verifiable instructions

Example: "Response must be under 200 words"

Curation: Resolved formatting inconsistencies in keyword arguments



Training Recipe

Dynamic Filtering

Removes samples where all rollouts are entirely correct or incorrect

Overlong Penalty

Penalizes samples failing to complete within max sequence length

Thinking Mode Only

Trained exclusively in thinking mode without reward model



Hyperparameters

Batch Size

128

Rollouts per Prompt

16

Learning Rate

3e-6

Training Steps

~180

Temperature

1.0

💡 IF-RL positioned as first stage because it can negatively impact human alignment, which subsequent RLHF stage recovers

Multi-Domain RL & RLHF

 Multi-Domain RL

MCQA STEM55%

Multi-choice question answering

Agentic Tool Calling30%

Workplace Assistant setup

Structured Output15%

Format adherence

70 steps | No performance degradation observed

 Why Multi-Domain?

☒ No degradation observed across evaluation benchmarks

☒ Consistent improvements on MMLU-Pro, τ^2 -Bench, IF-Bench

☒ Similar response lengths and verification times

 RLHF

Dataset

HelpSteer3 + arena-human-preference-140k + synthetic safety

GenRM

Qwen3-235B-A22B-Thinking-2507 as generative reward model

Training

Pair-wise comparisons with length-normalized reward

 RLHF Hyperparameters

Batch Size128

Rollouts16

Max Response16K

Learning Rate3e-6

KL Coefficient0.03

Steps~30

Long-Context RL & Code RL

Long-Context RL

Dataset

NVIDIA Nano-v3 RL blend, long-context only

32K

Input Tokens

49K

Max Sequence

Environment: Nemo-Gym with Qwen3-235B as LLM judge

Training: ~30 steps (stops when tokens increase rapidly)

Domain Isolation

Incorporating other domains during long-context RL **negatively affects performance** on unrelated benchmarks

Motivates **domain-specific training setup** for long-context capability

Code RL

Data Curation

3.5K high-difficulty prompts from competitive programming

Filtering

Remove prompts GPT-OSS-120B solves in all 8 rollouts

118K

Max Response

16

Rollouts

Reward: Strict binary to avoid hacking

Verification Infrastructure

Batch Size

Code executions per RL step

$128 \times 16 = 2,048$

CPU Cores

384

Batch Time

427.2s

Asynchronous reward verification server

Software Engineering RL (SWE RL)

Agentless RL

Data Source

Same as Wang et al. (2025) for code repair

Reward Model

GPT-OSS-120B evaluates code repair quality

128×16

Batch

98K

Max Seq

Filtering: Mask loss if no rollout gets reward > 0.5

Agentless RL Results

Agentless Mini avg@4

41.9% → 44.3%

OpenHands avg@4

49.8% → 50.8%

Generalizes to agentic settings, improving both agentless and agentic performance

Execution-Based Agentic RL

Scaffold

OpenHands framework with tool usage

Environment

Fully executable with compilation and test execution

1024

Batch (16×64)

256K

Context

200

Max Turns

0.8

Temperature

Training Data

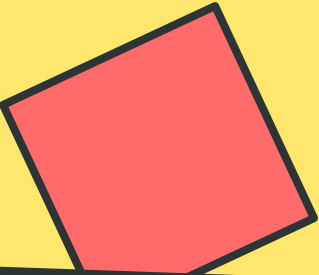
Sources

SWE-Gym + R2E-Subset

Filtering

Remove 100% accuracy (too easy), discard 90% of 0% accuracy (too hard)

Reward: Deterministic from compilation errors, failing tests, successful passes



Chapter 06

Multi-Domain On-Policy Distillation

Recovering capabilities and unifying expertise



MOPD Objective & Training Dynamics



Motivation

Cascade RL **reduces but doesn't eliminate** catastrophic forgetting

Noticeable fluctuations across benchmark categories during training

Dominant trade-offs differ by stage (e.g., RLVR reduces entropy, RLHF trades off instruction following)



Why MOPD Works

- 1 **Teacher selection:** Choose strongest validation checkpoint per domain
- 2 **Same initialization:** Teachers share tokenizer and vocabulary
- 3 **Dense token-level advantage:** More efficient than sparse outcome rewards



MOPD Objective

Token-level distillation advantage:

$$a_{\text{MOPD}} = \log \pi_{\text{teacher}}(y_t | s_t) - \log \pi_{\text{train}}(y_t | s_t)$$

Truncated importance weighting:

$$w_t = \text{sg}[r_t] \cdot 1[\epsilon_{\text{low}} \leq r_t \leq \epsilon_{\text{high}}]$$

Bounds:

$$\epsilon_{\text{low}} = 0.5, \epsilon_{\text{high}} = 2.0$$



Domain Teachers



Math Teacher

Initial SFT checkpoint (strong reasoning)



RLHF Teacher

Checkpoint after RLHF from SFT



Multi-domain Teacher

After IF-RL + Multi-domain RL

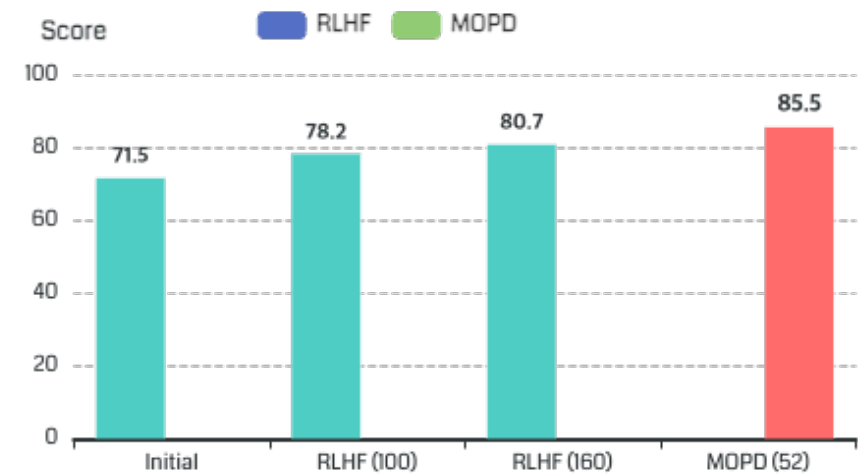
MOPD Training Efficiency

AIME 2025: MOPD vs GRPO



GRPO: 89.9 → 91.0 in 25 steps | **MOPD:** 89.9 → 92.0 in 30 steps (reaches teacher level)

ArenaHard v2: MOPD vs RLHF



Efficiency Advantage

✓ Dense Token-Level Advantage

MOPD provides per-token signal vs GRPO's sparse sequence-level reward

✓ Faster Convergence

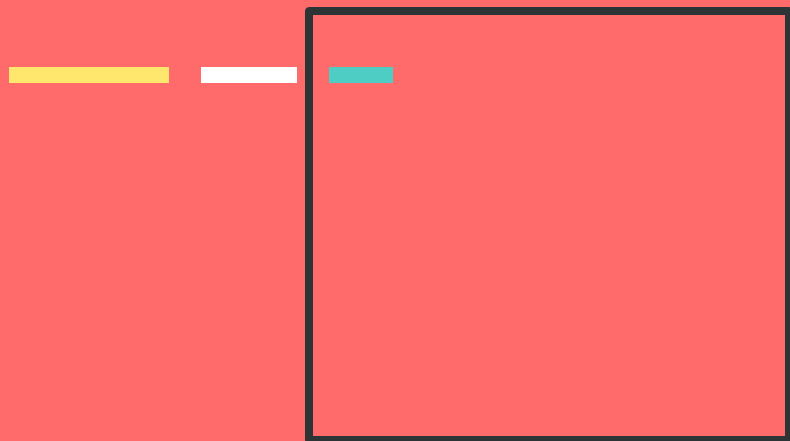
MOPD reaches stronger performance in fewer optimization steps

Example: MOPD achieves 85.5 on Hard Prompt in 52 steps vs RLHF's 80.7 in 160 steps

Chapter 07

IMO 2025 Results

Gold medal performance with test-time scaling



IMO 2025 Competition Results


35/42

Points

GOLD MEDAL

International Mathematical Olympiad 2025



Test-Time Scaling

Framework

Self-improving generate-verify-refine pipeline

Process

Iteratively generate candidates, verify, refine based on feedback

Evaluation: Human expert (IMO 2015 Gold medalist)

Problem Breakdown

P1	Problem 1 Geometry	7/7
P2	Problem 2 Analytic Geometry	7/7
P3	Problem 3 Combinatorics	7/7
P4	Problem 4 Number Theory	7/7
P5	Problem 5 Algebra	7/7
P6	Problem 6 Geometry (Hard)	0/7

Remarkable: 30B-A3B model solves 5/6 IMO problems, demonstrating strong olympiad-level reasoning with effective inference-time scaling

IMO-Proof Bench Performance

 Overall Performance

72.9%

IMO-Proof Bench

90+
Basic (30)

53.4
Advanced (30)

Within 8 points of DeepSeek-Math-V2-671B-A37B despite 10× fewer active parameters

 Test-Time Compute Scaling

Round 1	40.7%
Round 3	50.5%
Round 5	53.4%

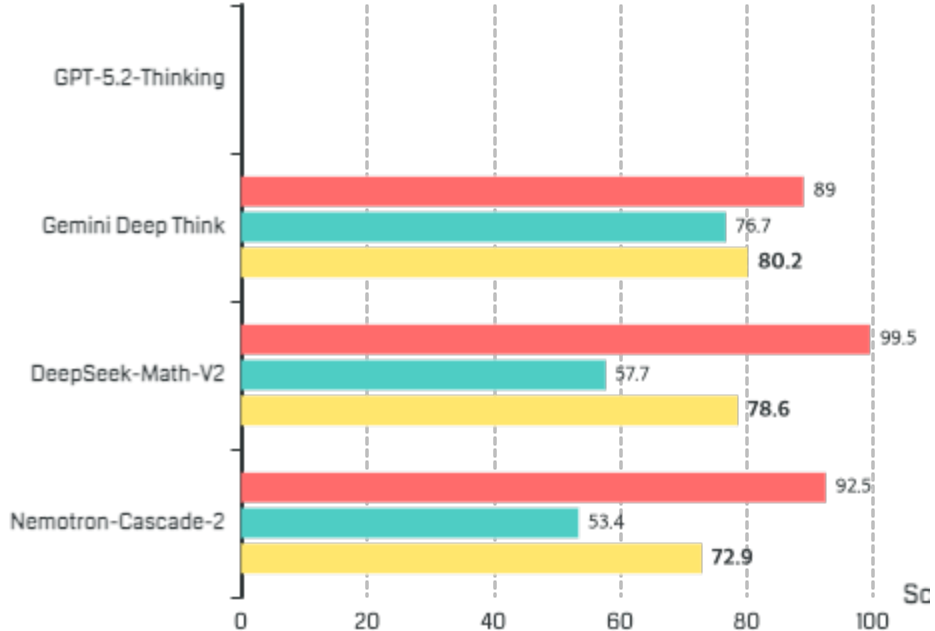
+12.7 points improvement from round 1 to round 5 on Advanced split

Comparison with Baselines

Basic (30)

Advanced (30)

Overall (60)



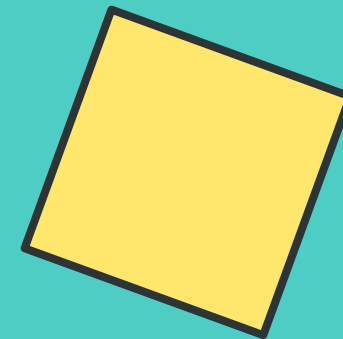
Model	Basic (30)	Advanced (30)	Overall (60)
GPT-5.2-Thinking	89	76.7	80.2
Gemini Deep Think	99.5	57.7	78.6
DeepSeek-Math-V2	92.5	53.4	72.9
Nemotron-Cascade-2			

Grading: DeepSeek-V3.2-Speciale as judge with LLM ProofAutoGrader prompt for reproducibility

Chapter 08

Competitive Coding Performance

Gold medal achievements in IOI and ICPC



IOI 2025 & ICPC World Finals 2025



IOI 2025 Gold Medal

439.28

/600 Points

10/12

Subtasks Full

≤40

Rounds

≤2000

Generations

Test-Time Scaling: Multi-round generate-select-submit framework



ICPC World Finals 2025

10/12

Problems Solved

#4

Placement

Gold

Medal

8 problems solved within 100 submissions (except A and I)



IOI Key Achievements

Full score on Problem 3 and Problem 4

86+ points on Problem 2 in just 5 rounds

Cross-subtask insights: Shared learning between subtasks



ICPC Methodology

Generation

Up to 1000 solutions per problem

Filtering

Initial filtering before submission

Evaluation

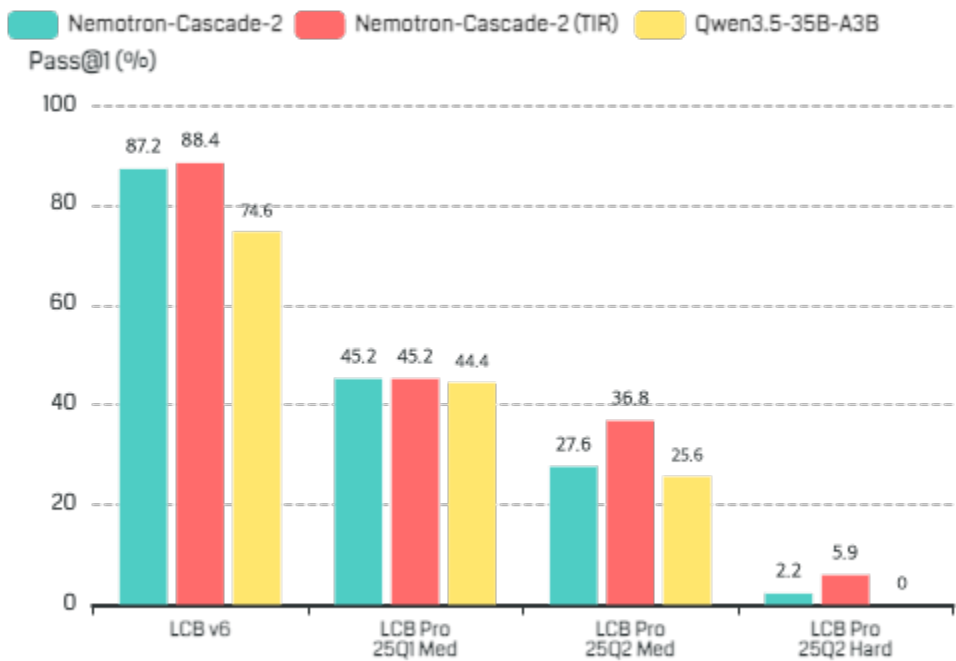
Official judge with test cases



Both competitions use official rules and evaluation - IOI with online judge, ICPC with official test cases

LiveCodeBench & Codeforces Performance

LiveCodeBench Results



87.2%

v6 Pass@1

88.4%

v6 with TIR



Codeforces ELO Rating

2345

ELO Rating

99.7%

Percentile (40 Div.1/Div.2 Rounds)



Competitive Advantages

Outperforms frontier open-source models with 100B+ total params

With TIR: Matches models with 300B+ params (Kimi-K2.5-1T, Qwen-3.5-397B, DeepSeek-v3.2-Speciale)

Hard problems: >0% on LiveCodeBench Pro Hard split within 8 attempts



Chapter 09

Training Hyperparameters

Complete configuration for reproducibility



SFT & Cascade RL Hyperparameters

SFT Configuration

Global Batch Size	64
Packed Sequence	256K
Learning Rate	5e-5 → 5e-6
Training Steps	33,000
Optimizer	AdamW
Scheduler	Cosine

IF-RL Configuration

Max Response	49K
Batch Size	128
Rollouts	16
Learning Rate	3e-6
Steps	~180

Multi-domain RL & MOPD

Multi-domain RL

Max Response: 49K	Batch: 128
Rollouts: 16	LR: 3e-6
Steps: ~70	Overlong: True

MOPD

Max Response: 98K	Batch: 128
Rollouts: 4	LR: 3e-6
Steps: ~52	ε: [0.5, 2.0]

RLHF Configuration

Max Response	16K
Batch Size	128
Rollouts	16
KL Coefficient	0.03
Steps	~25

Long-Context, Code & SWE RL Hyperparameters



Long-Context RL

Max Response 49K

Batch Size 128

Rollouts 16

Learning Rate 3e-6

Steps ~30

Overlong Filter True

Judge: Qwen3-235B for QA evaluation



Code RL

Max Response 118K

Batch Size 128

Rollouts 16

Learning Rate 3e-6

Steps ~22

Top-p 0.95

Reward: Strict binary (avoid hacking)



SWE RL

Agentless

Batch: 128×16
Max Seq: 98K
Steps: 40-50

Agentic (Exec-based)

Batch: 16×64
Context: 256K
Max Turns: 200
Temp: 0.8

Data: SWE-Gym + R2E-Subset with filtering

i All RL stages use GRPO with on-policy training, group-normalized rewards, and token-level loss

Key Takeaways



Efficient Post-Training

Cascade RL + MOPD achieves **frontier-level reasoning** with 20× fewer parameters



Gold Medal Performance

IMO 2025, IOI 2025, and ICPC World Finals **gold medals** with 30B-A3B model



Open Source Release

All artifacts released: model checkpoints, SFT data, RL data, and methodology



Research Impact

Enables **reproducibility, analysis, and extension** of the Cascade RL paradigm

30B MoE

3B Active

Gold Medal

Nemotron-Cascade 2

Post-Training LLMs with Cascade RL and Multi-Domain On-Policy Distillation

NVIDIA Research | March 2026